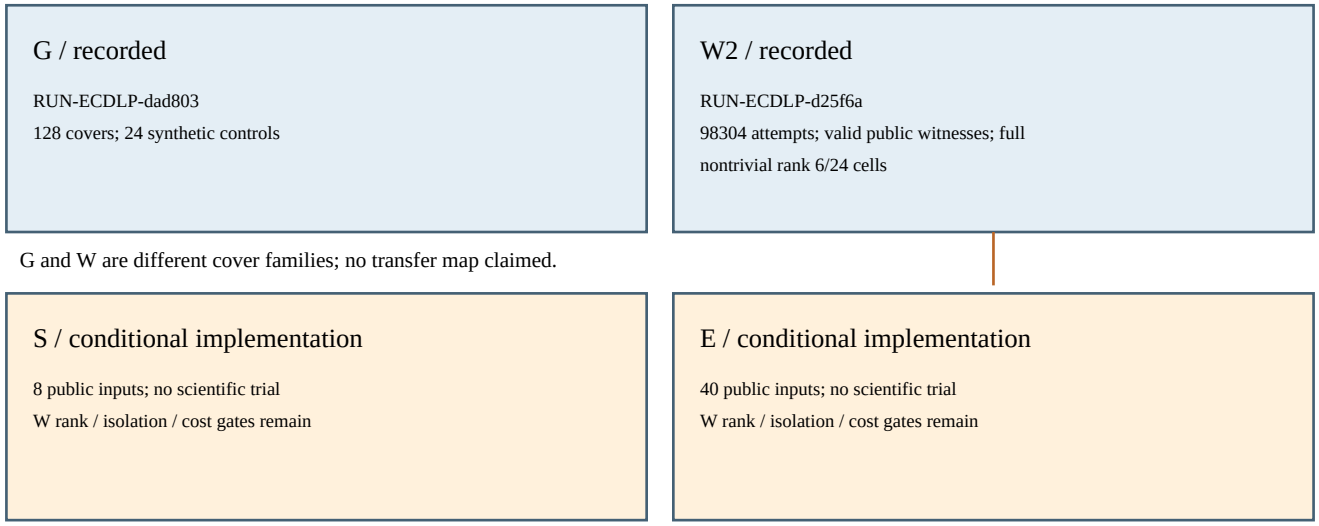
Prym: implementation and observations

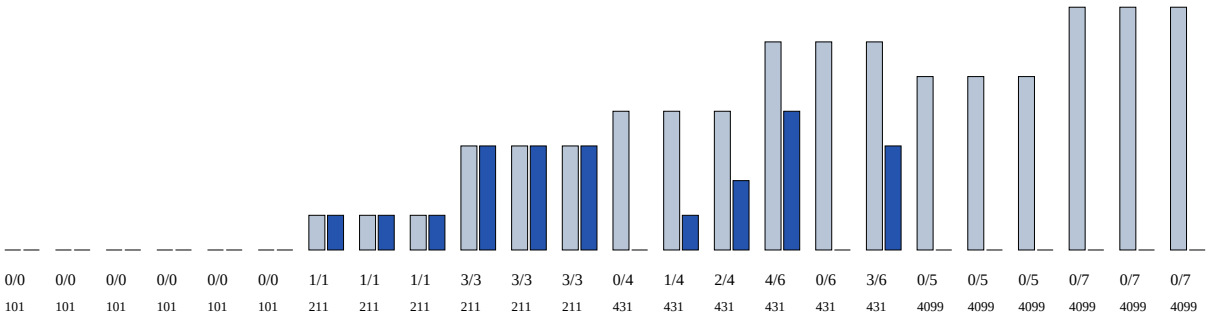
2026-10-21 / issue #2414 / PR #2415 / IDEA-20260909-4c5ee9. Observations and conditional software only; no scientific-state promotion or measured attack-cost advantage.

G: RUN-ECDLP-dad803 completed 64 primary and 64 companion covers plus 24 synthetic controls. Independent checker passed. Recorded Frobenius overlap: 3/64 primary, 4/64 companions. Only Fp/Fp2 traces were checked; no complete genus-3 certificate or Jacobian map.

W2: RUN-ECDLP-d25f6a completed 98,304 fixed attempts on eight curves and three within-curve seeds, with 11 controls. Independent source/divisor and rank checker passed. Original frozen W files remain intact; W2 corrects a manifest schema omission using fresh IDs.



RUN-ECDLP-d25f6a: measured rank / unknown columns; exact finite matrices



Grey bars are U (actual unknown norm columns); blue bars are rank. Values are exact, n=24 cells; seeds are not independent curves. Six cells attain rank U>0; six have U=0 from the public source schedule; twelve miss U. All F4099 cells accepted zero rows. This limits the tested finite stream, not all Prym algorithms or deployed curves.

Source/rank observations: experiments/EXP-ECDLP-cded08/runs/RUN-ECDLP-d25f6a/{raw-result.json,verification.json}; geometry: experiments/EXP-ECDLP-da4ba3/runs/RUN-ECDLP-dad803/. Exact commands, resources, dependencies and inspected per-run reports are linked from README.md.

Conditional follow-ups and release gates

Integrated S software: canonical finite train/holdout streams, frozen histogram models, real/permuted/split-product/coordinate arms, identity A/A, independent full-pool algebraic census, and isolated workers. Complete geometric-orbit/stream and cost auditing still need release.

Integrated E software: same-base IC comparator, explicit target descent, salted challenge commitment and private verifier, exact attack-output seals, independent final source check, coordinate/null arms and finite plain rho. The concrete mitm m=2 comparator is not qualified as current best; the plain rho is not a certified optimized usable-action reference.

Public inputs: eight fresh S curves at p=4099,8191,16381,32749; forty E instances with subgroup bits 11,13,15,17. All search rejections are retained. Field/order/generator/base checks passed independently. EC1 identities accompany inputs; no E secrets or point-log labels were supplied.

S prospective revision: at p=101 the genus-2 Hasse bound allows at most 125,440 canonical candidates in the old full-source policy, less than the requested 131,072 distinct train-plus-holdout candidates. The revised fresh panel and 256-source rule precede all S labels; prior W2 outcomes are disclosed. Exhaustion still must be retained.

Isolation is unavailable in this runtime: Landlock ENOSYS and bubblewrap user-ID mapping failure. Workers refuse, without an unisolated fallback. The software test proves refusal here, not successful isolation. Scientific S and E have not executed and remain review_required.

Thirteen planted software controls pass: interpolation/census, malformed witnesses, rank, aliases, A/A/coordinate transport, finite support, V2 manifest schema, same-base IC/planted descent, seal mutations, gate and isolation refusal. These are correctness tests, not blind recovery or performance evidence.

Next gates: prospectively frozen nontrivial full-rank constructor or distinct repair; enforceable worker isolation; independently qualified current IC, complete S weights and reference protocols; full native plan/handoff/archive/admission. Raw vectors exclude pow internals, Python bookkeeping and independent checking; no complete S or end-to-end gain is inferred.

Sources: README.md, status.json, input-validation.json, software-tests.txt, diagram-source.json and recorded-ranks.json. Original design report is historical. No canonical scientific frontier graph changes follow from this observational report. PDF inspected before publication.